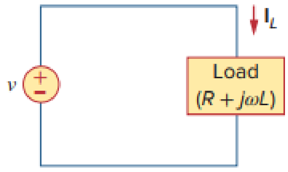
Chapter 9, Problem 32P

Problem

Using Fig. 9.40, design a problem to help other students better understand phasor relationships for circuit elements.

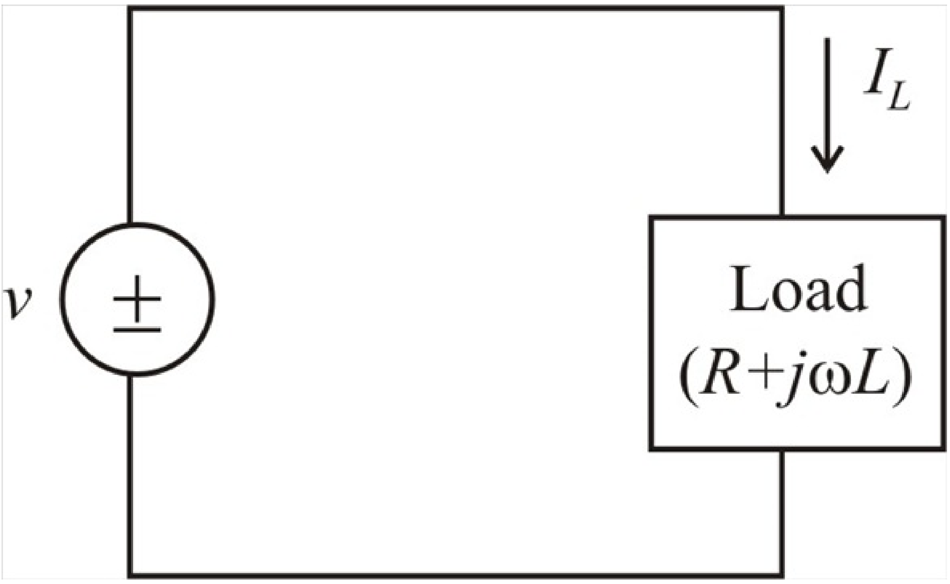
Figure 9.40:



Step-by-step solution

Step 1 of 2

Given circuit ,



Step 2 of 2

From the above circuit ,

$$Z = R + j\omega L$$

Hence , the circuit

$$I_L = \frac{v}{Z}$$

$$I_L = \frac{v}{R + j\omega L}$$

Convert rectangular form to polar form

$$I_L = \frac{v}{\sqrt{R^2 + (\omega L)^2} \angle \tan^{-1}\left(\frac{\omega L}{R}\right)}$$

$$\therefore \boxed{I_L = \frac{v}{\sqrt{R^2 + (\omega L)^2}} \angle\left(-\tan^{-1}\left(\frac{\omega L}{R}\right)\right)}$$